

RESEARCH DESIGN



RESEARCH DESIGN

2nd step in a research study

Three types

- Randomized experiment (true experiment)
- Quasi-experiment (non-randomized experiments)
- Observational Design (non-experimental)

RESEARCH DESIGN: RANDOMIZED EXPERIMENT

Definition:

- a research technique that ***attempts to*** establish the
 - causal relationship between an independent variable (x) and a dependent variable (y) by
 - randomly assigning participants to experimental groups
 - characterized by differing levels of x , and
 - measuring the average behavior y that results in each group.

RESEARCH DESIGN: EXPERIMENTAL METHOD

Random Assignment equates the groups on average

Underlying Idea: Test differences between groups with statistical tests

to determine if the difference is larger than would be expected by chance



RESEARCH DESIGN

Randomized Experiment (strong research design)

- The only design that strongly supports causal inference

Two things are related (correlated)

- A caused B
- B caused A
- C caused both A and B
 - Names for C
 - Lurking Variable,
 - Confounding Variable,
 - Third Variable
 - Spurious Relationship



C CAUSED BOTH A AND B

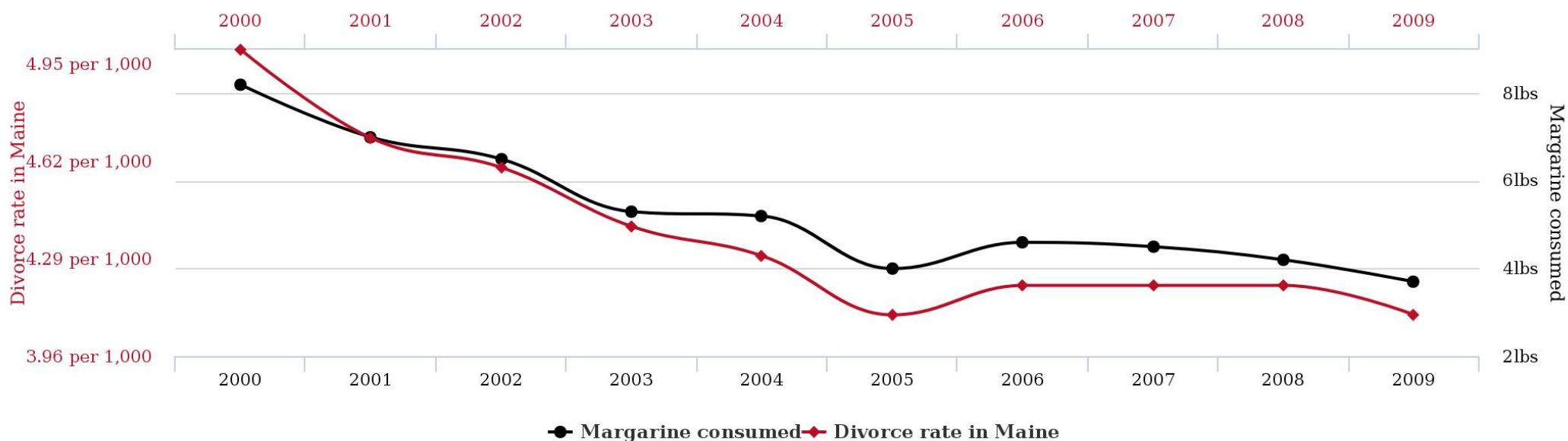


C CAUSED BOTH A AND B

Divorce rate in Maine

correlates with

Per capita consumption of margarine



tylervigen.com

<http://tylervigen.com/spurious-correlations>

ACTIVITY

Cities with a more police officers have more crimes

Elementary school children with larger shoes have higher test scores.

Number of storks sighted and the population of Oldenburg, Germany

Teachers' salaries and the price of Starbucks coffee

Tea drinkers have lower rates of lung cancer

SIMPSON'S PARADOX



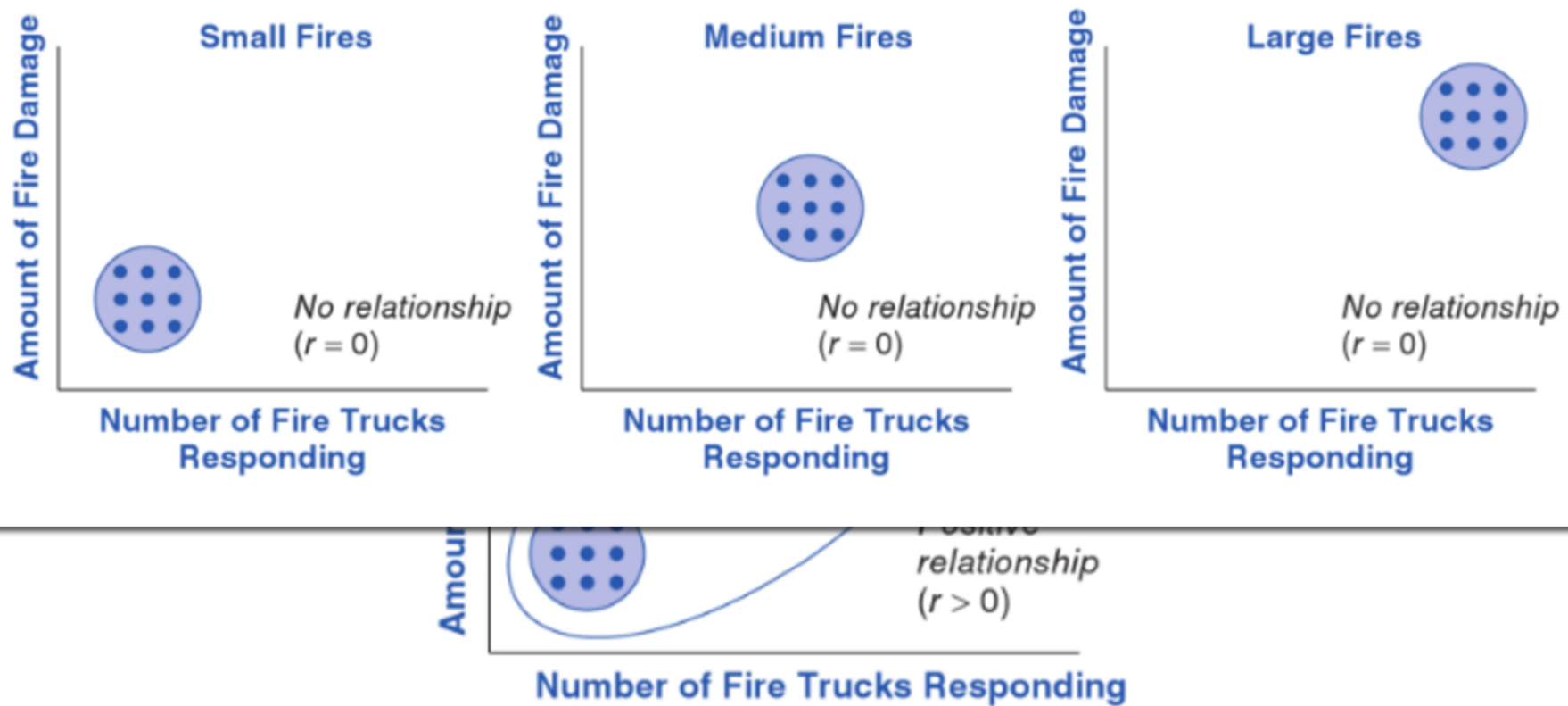
SIMPSON'S PARADOX

Is a trend appears in different groups of data but disappears or reverses when these groups are combined



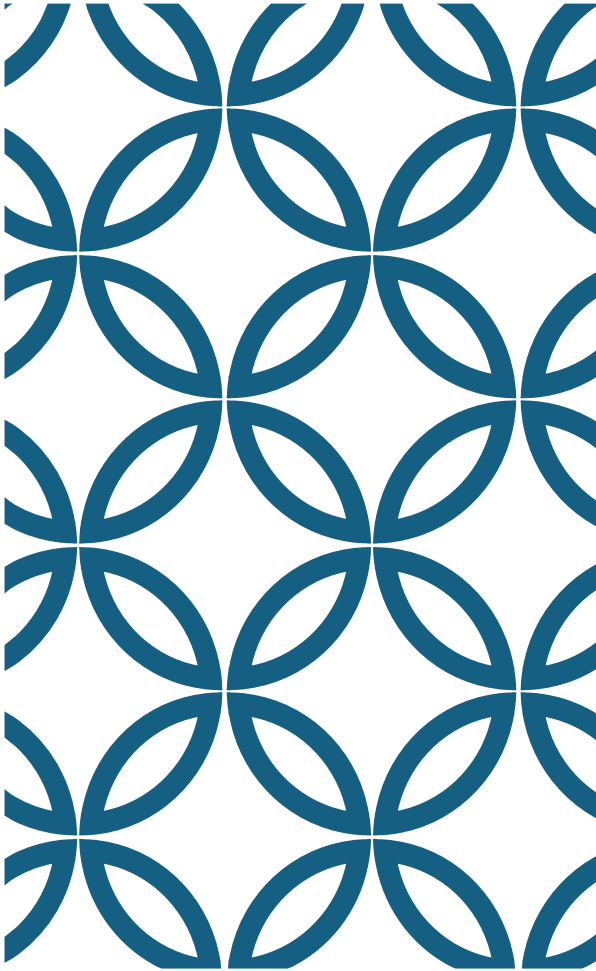
SIMPSON'S PARADOX

(b) After controlling for size of fire





WRAPPING UP...



PHILOSOPHICAL BACKGROUND

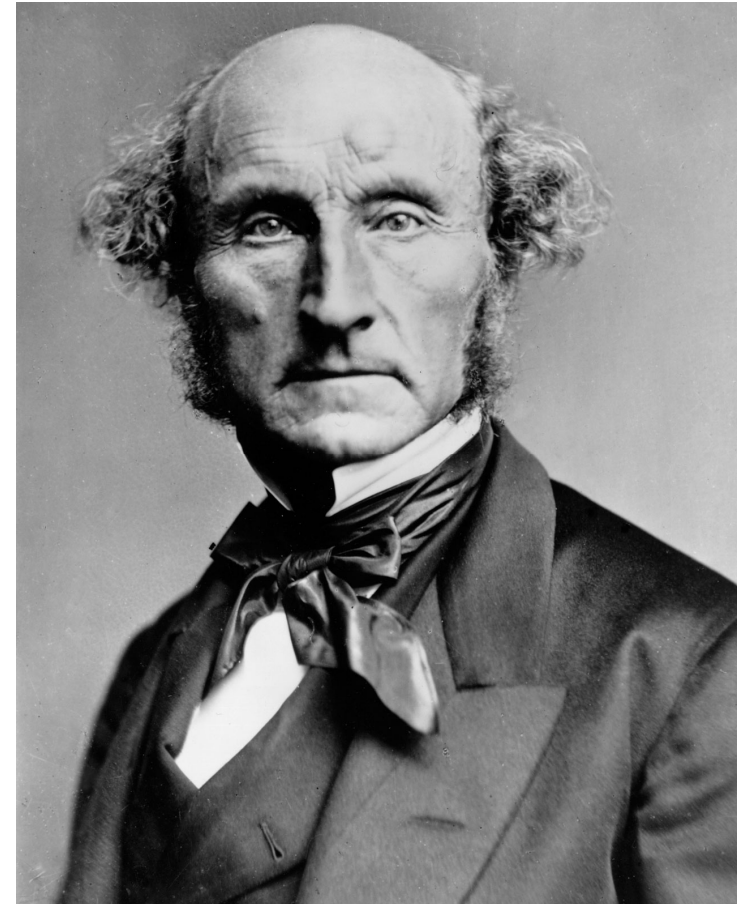
BACKGROUND AND HISTORY

John Stuart Mill wrote

- “A system of logic, ratiocinative and inductive, being a connected view of the principles of evidence and the methods of scientific investigation.”(Mill 1846)

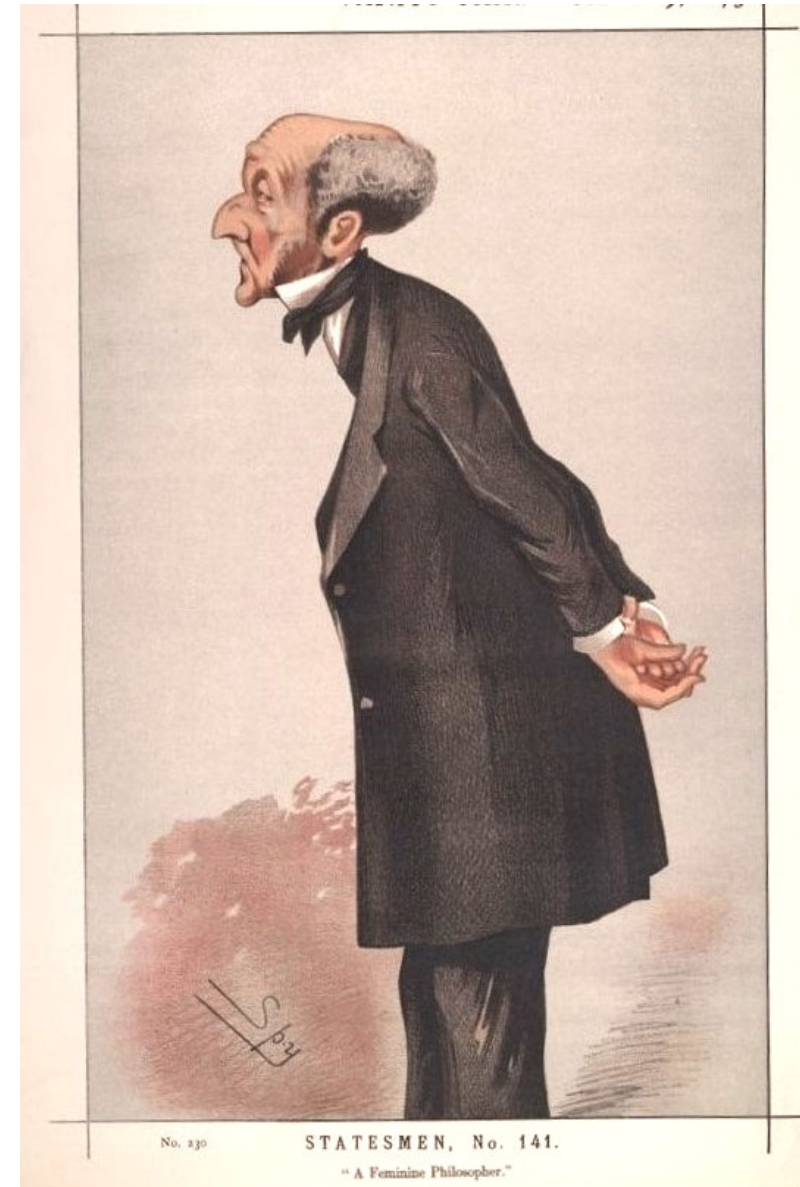
He Developed the Logic of Causation

- Chapter 8, Of the Four Methods of Experimental Inquiry
- Five Canons (Methods of Experimental Inquiry)
 - Direct method of agreement
 - Method of agreement and difference
 - Method of residue
 - Method of concomitant variations
 - Method of Differences



J. S. MILL

- Five Canons (Methods of Experimental Inquiry)
 - Direct method of agreement
 - Method of agreement and difference
 - Method of residue
 - Method of concomitant variations
 - Method of differences



Five Canons

First Canon: Method of Agreement

If two or more instances of the phenomenon under investigation have only one circumstance in common, the circumstances in which alone all the instances agree is the cause (or effect) of the given phenomenon.

Second Canon: Method of Difference

If the instance in which the phenomenon under investigation occurs and an instance in which it does not occur, have every circumstance in which alone the two instances differ is the effect, or the cause, or an indispensable part of the cause, of the phenomenon.



Five Canons

Third Canon: Joint Method of Agreement and Difference

If two or more instances in which the phenomenon occurs have only one circumstance in common, while two or more instances in which it does not occur have nothing in common save the absence of that circumstance, the circumstance in which alone the two sets of instance differ is the cause of the phenomenon.

Fourth Canon: Method of Residues

Subduct from any phenomenon such part as is known by previous inductions to be the effect of certain antecedents and the residue of the phenomenon is the effect of the remaining antecedents.



Five Canons

Fifth Canon: Method of Concomitant Variation

Whatever phenomenon varies in any manner whenever another phenomenon of the particular manner, is either a cause or an effect of that phenomenon or is connected with it through some fact of causation.



BACKGROUND AND HISTORY



Neyman and Fisher (1920s/1930s)

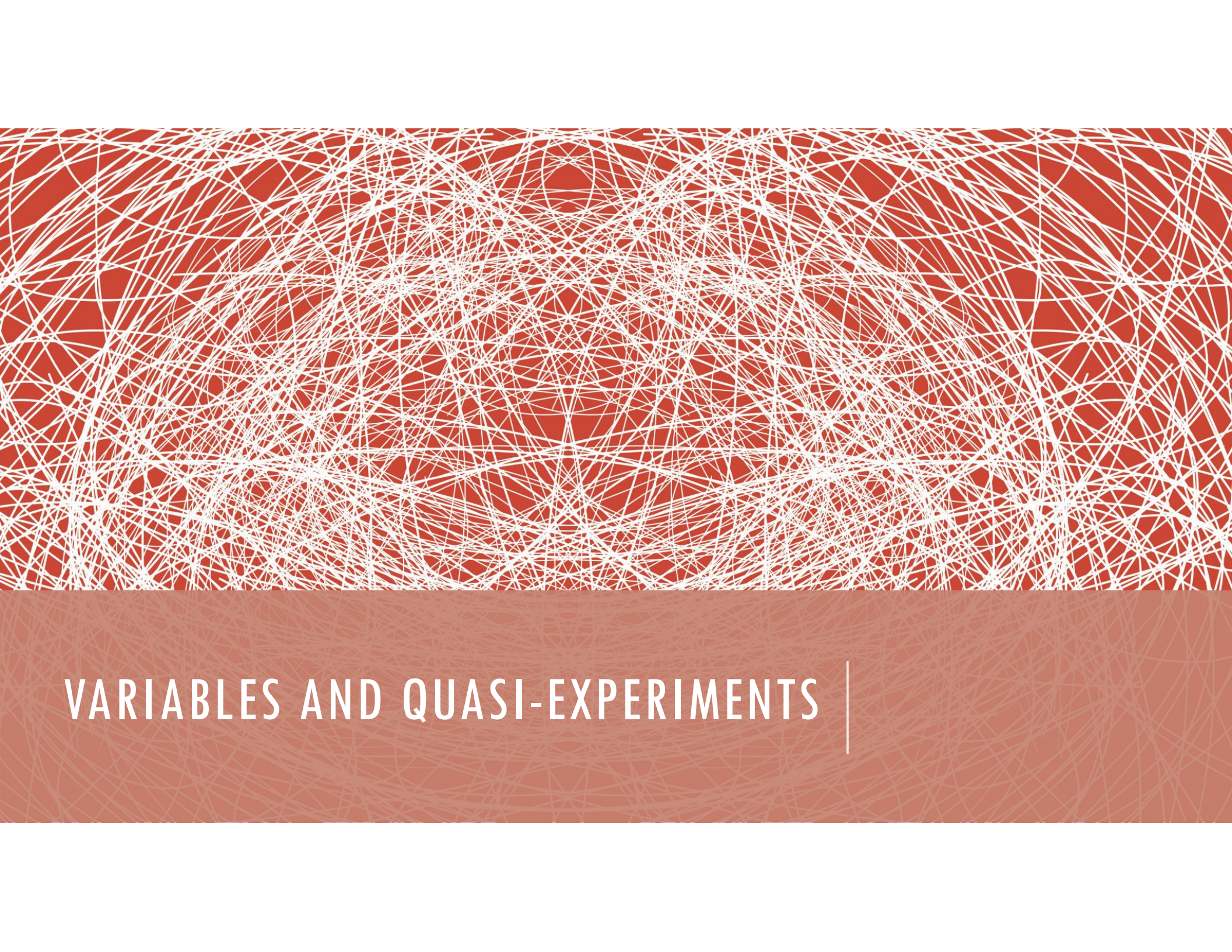
- Jerzy Neyman (1923)
 - In experiments
 - In survey sampling (1934)
 - Neyman, J.(1934) "On the two different aspects of the representative method: The method of stratified sampling and the method of purposive selection", *Journal of the Royal Statistical Society*, 97 (4), 557–625
- Ronald Fisher (1935)
 - *The Design of Experiments*

Random assignment

- equates groups on all confounding variables ON THE AVERAGE (across replications)
- Treatment Group
- Control Group

WRAPPING UP...





VARIABLES AND QUASI-EXPERIMENTS

VARIABLES

Definition: A feature of an individual that can take on different values

Conceptual Variable

- Def: Researcher's specification of the topic of study

Operational Definition

- Def: Links the conceptual variable to one that can be measured.

The image shows a chalkboard with handwritten mathematical derivations. On the left, the text "Secant Lines" is written next to a graph of a curve $y = g(x)$. On the right, the limit process for finding the derivative is shown:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$
$$f(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$
$$= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$
$$= \lim_{h \rightarrow 0} h(2x + h)$$

The final result is $2x$.

VARIABLES



Explanatory Variable

Def: variable measured or manipulated by the researcher; the possible cause
Same as independent variable



Response Variable/ Dependent Variable

Def: outcome of interest



Lurking / Confounding Variable

Def: Uncontrolled variable that hides the relationship between the explanatory and the response variable.

ACTIVITY

How would you test these relationships using Random Assignment?

- Cities with a more police officers have more crimes
- Elementary school children with larger shoes have higher test scores.
- Number of storks sighted and the population of Oldenburg, Germany
- Teachers' salaries and the price of Starbucks coffee
- Tea drinkers have lower rates of lung cancer
- Shark attacks and Ice Cream sales

QUASI-EXPERIMENTAL DESIGN

Non-random experiment

Def. Has all the features of a randomized experiment, except random assignment

Reasons why we can't do a randomized experiment?



QUASI-EXPERIMENTAL DESIGN

Reasons why we can't do a randomized experiment?

- May be logically impossible because subjects show up with their explanatory variable features defined
- Ethically problematic
- Impractical

We worry more about threats to validity

QUASI-EXPERIMENTAL DESIGN

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 - Can't withhold treatment
 - Can't inflict harm
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QUASI-EXPERIMENTAL DESIGN

Reasons why we can't do a randomized experiment?

- May be logically impossible because subjects show up with their explanatory variable features defined
- Ethically problematic
 - Can't withhold treatment
 - Can't inflict harm
- Impractical
 - Too long for data collection
 - Too expensive to collect data
 - 1 time event

We worry more about threats to validity

EXAMPLE: TWIN STUDIES

Twins

- Monozygotic Twins Share 100% of their DNA.
- Dizygotic Twins Share 50% (like regular siblings).

Can be used to untangle the influences on characteristics by

- genes, and the
- environment
 - by comparing twins and twin types.



OTHER USES FOR TWINS: NASA TWINS

10 investigations were conducted with identical twin astronauts [Scott](#) and [Mark Kelly](#).

- Physiology,
- Behavioral Health,
- Microbiome,
- Molecular/Omics

Provide broader insight into the

- subtle effects and changes that may occur in spaceflight as compared to Earth by studying two individuals who have the
 - same genetics, but are in
 - different environments for [one year](#).



EXAMPLE: TWIN STUDIES

IDENTICAL TWINS (NO SURGERY)



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EXAMPLE: TWIN STUDIES





WRAPPING UP...